

1. Which of the following is used to add styles and formatting to a webpage in CSS?

- a. Functions
- b. Variables
- c. Selectors
- d. Arrays

Correct answer: c. Selectors

2. Which one of the following techniques is not used in the Binary tree?

- a. Randomized traversal
- b. Preorder traversal
- c. Postorder traversal
- d. Inorder traversal

Correct answer: a. Randomized traversal

3. Which of the following principles does Queue use?

- a. LIFO principle
- b. FIFO principle
- c. Linear tree
- d. Ordered array

Correct answer: b. FIFO principle

4. Which of the following is not considered as a peripheral device?

- a. CPU
- b. Keyboard
- c. Monitor
- d. All of the above

Correct answer: a. CPU

5. Which recovery technique uses the undo/redo approach?

- a. Immediate Update
- b. Deferred Update
- c. ARIES Algorithm
- d. None

Correct answer: a. Immediate Update

6. Which of the following statements is false about classes of language in automaton theory?

- a. Context-free language is the subset of context-sensitive language
- b. Regular language is the subset of context-sensitive language
- c. Recursively enumerable language is the superset of regular language
- d. Context-sensitive language is a subset of context-free language

Correct answer: d. Context-sensitive language is a subset of context-free language

7. Where are the list of processes that are prepared to be executed and waiting placed?

- a. Job queue
- b. Ready queue
- c. Execution queue
- d. Process queue

Correct answer: b. Ready queue

8. Which one of the following statements is not true about the application of micro-operations?

- a. Selective set operation is implemented using OR micro-operation
- b. Selective clear operation clears to zero the bit in the first operand only where there are corresponding 1's in the logical operand
- c. Clear operation is implemented using Exclusive-OR Micro-operation.
- d. None of the above

Correct answer: d. None of the above

9. Function MY_AGENT (PERCEPT) return action

static state, a description of the current world state

goal, a description of the goal to achieve may be in terms of state

state $\leftarrow$ UPDATE_STATE(state, percept)

actionSet $\leftarrow$ POSSIBLE_ACTIONS(state)

action $\leftarrow$ BEST_ACTION(actionSet)

state $\leftarrow$ UPDATE_STATE(state, action)

return action

Based on the above prototype, the instruction is used to:

- a. Modify the state of the environment if necessary
- b. Update the state information to be like the one after executing the best action chosen
- c. Update the environment so that the agent can perceive the next percept
- d. Modify the state of the environment once the decision of which action to choose is made.

Correct answer: a. Modify the state of the environment if necessary

10. The process of recovering the original message without the detailed knowledge of the secret information is called:

- a. Decryption
- b. Decipher
- c. Cryptanalysis
- d. A and B

Correct answer: c. Cryptanalysis

11. Both the CISC and RISC architectures have been developed to reduce the _____

- a. Time delay
- b. Semantic gap
- c. Cost
- d. All of the mentioned

Correct answer: b. Semantic gap

12. All of the following are disadvantages of using DHCP services EXCEPT?

- a. Fault on the DHCP server can affect the whole network
- b. It is not a secured way of assigning IP addresses
- c. Static IP address assignment can't be used.
- d. It minimizes the load of network admin

Correct answer: d. It minimizes the load of network admin

13. CPU scheduling is the basis of _____

- a. Multiprogramming operating systems
- b. Larger memory-sized systems
- c. Multiprocessor systems
- d. None of the mentioned

Correct answer: a. Multiprogramming operating systems

14. What is the bottom-up parsing method also known as?

- a. Recursive descent parsing
- b. Predictive parsing
- c. Shift-reduce parsing
- d. All of them

Correct answer: c. Shift-reduce parsing

15. On the basis of the general search methods, which simple strategy results in greedy best search?

- a. Append newly generated nodes at the beginning of the open list
- b. Append newly generated nodes at the end of the open list
- c. Insert newly generated nodes in the open list according to their path cost (lowest values first)
- d. Insert newly generated nodes in the open list according to their evaluation function (lowest values first)

Correct answer: d. Insert newly generated nodes in the open list according to their evaluation function (lowest values first)

16. What is the main goal of data compression while it paths in different network layers? Its goal is to:

- a. Securing data

- b. Minimize the size of data
- c. Select the best path of destination devices
- d. All of the above

Correct answer: b. Minimize the size of data

17. The _____ statement causes a loop to terminate immediately.
- a. Continue
 - b. Goto
 - c. Break
 - d. Jump

Correct answer: c. Break

18. Which of the following can be a heuristic for grouping objects into subsystems?
- a. Assign objects identified in one use case into the same subsystem.
 - b. Create a dedicated subsystem for objects used for moving data among subsystems.
 - c. Minimize the number of associations crossing subsystem boundaries.
 - d. All of the above

Correct answer: d. All of the above

19. The term Push and Pop is related to _____
- a. Queue
 - b. Stack
 - c. Both
 - d. None

Correct answer: b. Stack

20. Which one of the following statements is false?
- a. A sentence is said to be invalid iff it is unsatisfiable.
 - b. A sentence is valid if it is satisfiable.
 - c. A sentence which is not invalid is satisfiable.
 - d. A sentence is unsatisfiable if it is false in all models.

Correct answer: b. A sentence is valid if it is satisfiable.

21. Which of the following PHP code snippets is used to retrieve data from a MySQL database?
- a. `$result = mysql_query("SELECT * FROM users WHERE id = 1");`
 - b. `$result=mysqlquery(conn, "SELECT * FROM users WHERE id = 1");`
 - c. `$result = $pdo->query("SELECT * FROM users WHERE id = 1");`
 - d. `$result = pg_query("SELECT * FROM users WHERE id = 1");`

Correct answer: `$result=mysqlquery(conn, "SELECT * FROM users WHERE id = 1");`

22. Which of the following JavaScript code snippets is used to add a new element to an HTML document?

- a. `element.appendChild(document.createTextNode("Hello, world!"));`
- b. `var element = document.createElement("div");`
- c. `document.body.appendChild(element);`
- d. `document.getElementById("container").appendChild(element);`

Correct answer: `var element = document.createElement("div");`

23. Consider the following function prototype to answer the following consecutive two questions below:

```
Function MY_AGENT (PERCEPT) return action
static state, a description of the current world state
goal, a description of the goal to achieve may be in terms of state
stateβUPDATE_STATE(state, percept)
actionSetβPOSSIBLE_ACTIONS(state)
actionβBEST_ACTION(actionSet)
stateβUPDATE_STATE(state,action)
return action
```

Based on the above prototype, MY_GENT is _____ type of agent.

- a. Goal based
- b. Model based reflex
- c. Simple reflex
- d. Utility based

Correct answer: Utility based

24. Which of the following statements is true about crackers?

- a. They attempt to break the system without permissions
- b. They are legally employed to break into the system
- c. They reveal the information after breaking into the system
- d. All

Correct answer: a. They attempt to break the system without permissions

25. Which one of the following is not the application of the stack data structure?

- a. String reversal
- b. Recursion
- c. Backtracking
- d. Asynchronous data transfer

Correct answer: d. Asynchronous data transfer

26. Which of the following is false about subsystems?

- a. A physical and replaceable part of the system.
- b. It reduces the complexity of the application domain.
- c. Decompose a system into simpler parts.

d. Well-defined interfaces that encapsulate the state and behavior of its contained classes.

Correct answer: It reduces the complexity of the application domain.

27. Suppose you are asked to construct a DFA for the strings ending with an 'n' length substring. How many minimal states may the DFA require?

- a. n
- b. n+1
- c. n+2
- d. n*2

Correct answer: n+1

28. Which type of firewall can filter incoming signals based on port numbers and protocol?

- a. Packet Filtering
- b. Proxy Service
- c. Stateful Inspection
- d. Threat-Focused NGFW

Correct answer: Stateful Inspection

29. Which of the following

is an example of Usability requirements?

- a. User interface
- b. Robustness
- c. Mean time to failure
- d. Adaptability

Correct answer: User interface

30. _____ is an alternative to log-based recovery.

- a. Disk recovery
- b. Shadow paging
- c. Disk shadowing
- d. Crash recovery

Correct answer: Shadow paging

31. To test the IP stack on your local host, which IP address would you ping?

- a. 127.0.0.1
- b. 1.0.0.127
- c. 127.0.0.0
- d. 127.0.0.255

Correct answer: 127.0.0.1

32. Why is determining the run time of a program exactly difficult in terms of algorithm performance?
- a. Processor speed
 - b. Current processor load
 - c. Input size of the given algorithm
 - d. All

Correct answer: All

33. Which one of the following is not the purpose of the super keyword?
- a. Refer to the current object
 - b. Call superclass constructor
 - c. Access hidden variables
 - d. Invoke superclass method with the same name as the subclass method

Correct answer: Refer to the current object

34. What will be the encrypted message if the message "ATTACK" is encrypted by Caesar's cipher with key = 25?
- a. ZSSZBJ
 - b. ZGGZXP
 - c. ZTTZAK
 - d. BGGBDL

Correct answer: ZSSZBJ

35. A translator that translates a single-line instruction at a time is:
- a. Compiler
 - b. Assembler
 - c. Interpreter
 - d. Linker

Correct answer: Interpreter

36. What is an operating system?
- a. Interface between the hardware and application programs
 - b. Collection of programs that manages hardware resources
 - c. System service provider to the application programs
 - d. All of the mentioned

Correct answer: All of the mentioned

37. Most of the time, it is recommended to apply a Hardware RAID into your system in terms of system performance. Why? Because Hardware RAID:
- a. A physical storage device that is built into a system
 - b. Can appear as a single hard disk in the system
 - c. There is no difference between regular hardware and the RAID devices
 - d. Can use its own resources

Correct answer: Can use its own resources

38. A conceptual level refers to:
- a. Describes the represents global view of the entire database
 - b. Describes the part of the database that a particular user group views of the data environment
 - c. Describes how data is actually stored on disk
 - d. Describes the physical storage structure of the database

Correct answer: Describes the represents global view of the entire database

39. Which of the following protocols in MTA enables users to access mail data/information from the MDA?
- a. POP3
 - b. IMAP
 - c. SMTP
 - d. Both POP3 and IMAP

Correct answer: Both POP3 and IMAP

40. Which one of the following has the same name as the class name?
- a. Method
 - b. Variable
 - c. Constructor
 - d. Package name

Correct answer: Constructor

41. An AI agent acts upon the environment using _____.
- a. Sensors
 - b. Effectors
 - c. Actuators
 - d. Manipulators

Correct answer: a. Sensors

42. Which of the following is used to include an external JavaScript file in an HTML document?
- a. `<script href="file.js"></script>`
 - b. `<link rel="stylesheet" href="file.js">`
 - c. `<script type="text/javascript" src="file.js"></script>`
 - d. `<link rel="stylesheet" type="text/javascript" href="file.js">`

Correct answer: c. `<script type="text/javascript" src="file.js"></script>`

43. How many tokens are there in the following C++ statement? `for (int i = 0; i <= 10; i++)`
- a. 4
 - b. 5
 - c. 9

d. 10

Correct answer: c. 9

44. Which of the following is not correct about a finite Automaton?

- a. Finite automaton accept a string or word by final state
- b. Finite Automaton has at least one final state
- c. Finite Automaton has at least one initial state
- d. Deterministic Finite Automaton has a single transition.

Correct answer: d. Finite Automaton has at least one initial state

45. Which of the following is used to define the structure of a webpage in HTML?

- a. Tags
- b. Classes
- c. IDs
- d. Styles

Correct answer: a. Tags

46. Which of the following ensures the atomicity of the transaction?

- a. Transaction management component of DBMS
- b. Application Programmer
- c. Concurrency control component of DBMS
- d. Recovery management component of DBMS

Correct answer: d. Recovery management component of DBMS

47. Suppose you want to design a database for a certain company after you know the need of all users of the database. What is the next step you are expected to do?

- a. Logical design
- b. Normalization
- c. Query Processing
- d. Conceptual design

Correct answer: d. Conceptual design

48. Which statement is true about the class concept?

- a. The role of a class is to define the attributes and methods.
- b. Class is used to distinguish one type of object from another.
- c. A set of objects that share common methods, structure, and behavior.
- d. All of the above

Correct answer: d. All of the above

49. Identify the data structure which has minimum access time in the case of symbol table implementation?

- a. Self-organizing list
- b. Hash table

- c. Search tree
- d. Linear

Correct answer: b. Hash table

50. The data link layer divides the stream of bits received from the physical layer into manageable data units called _____?

- a. Bits
- b. Frames
- c. Packets
- d. Segments

Correct answer: b. Frames

51. Which of the following is an example of a Dynamic Programming algorithm?

- a. Breadth First Search
- b. Depth First Search
- c. Dijkstra's Algorithm
- d. Knapsack Problem

Correct answer: c. Dijkstra's Algorithm

52. Consider the following outline of a nested if-else structure:

```
if (condition1)
{
    if (condition2)
        statement1;
    else
```

```
statement2;
```

```
}
```

Which of the statements below is true regarding this structure?

- a. Syntactically, it is invalid to have more if clauses than else clauses.
- b. Statement2 will only execute if condition1 is false and condition2 is false.
- c. Statement2 will only execute if condition1 is true and condition2 is false.
- d. Statement2 will only execute if condition1 is false, it does not matter what condition2 is.

Correct answer: c. Statement2 will only execute if condition1 is true and condition2 is false.

53. What should be considered when designing an algorithm?

- a. If the hardware is used correctly.
- b. If the software is used correctly.
- c. If there is more than one way to solve the problem.
- d. All of the above

Correct answer: c. If there is more than one way to solve the problem.

54. Which one is used to achieve the encapsulation concept in Java?
- a. Declare the variables of a class as private.
 - b. Provide public setter and getter methods to modify and view the variable's values.
 - c. By using an Interface.
 - d. A & B

Correct answer: d. A & B

55. Which one of the following is the collection of statements that performs a specific operation?
- a. Class
 - b. Package
 - c. Variable
 - d. Method

Correct answer: d. Method

56. Which of the following is used to create dynamic content and perform server-side processing in PHP?
- a. Functions
 - b. Variables
 - c. Classes
 - d. Objects

Correct answer: a. Functions

57. Which of the following is a commonly used JavaScript library for creating interactive web interfaces?
- a. jQuery
 - b. React
 - c. Angular
 - d. Vue.js

Correct answer: a. jQuery

58. Which of the following is an example of a Dynamic Programming algorithm?
- a. Breadth First Search
 - b. Depth First Search
 - c. Dijkstra's Algorithm
 - d. Knapsack Problem

Correct answer: c. Dijkstra's Algorithm

59. If a database server is referenced in a distributed transaction, the value of its commit point strength determines which role it plays in the _____.

- a. Two-phase commit
- b. Transaction locking
- c. Two-phase locking

d. Checkpoints

Correct answer: a. Two-phase commit

60. Which of the following CSS code snippets is used to create a responsive design for mobile devices?

- a. `@media screen and (min-width: 768px) { ... }`
- b. `.container { display: flex; }`
- c. `box { transform: rotate(45deg); }`
- d. `.menu { background-color: #foo; }`

Correct answer: b. `.container { display: flex; }`

Sure! Here's the organized version of the questions:

61. Why is query processing and optimization important in Database?

- a. Both are used to process SQL statements to return answers.
- b. Both help the user retrieve results from the database based on time.
- c. Both improve the performance of the database by efficiently executing a query.
- d. Both are important in executing queries to access information from the DB.

The correct answer is: Both improve the performance of the database by efficiently executing a query.

62. In the linked list implementation of a queue, where will the new element be inserted?

- a. At the middle position of the linked list.
- b. At the head position of the linked list.
- c. At the tail position of the linked list.
- d. None of the above.

The correct answer is: At the tail position of the linked list.

63. Among the following states, which would be denoted as the final state/acceptance state?

$L = \{x \in \Sigma^* \mid \text{length of } x \text{ is at least } 2\}$

- a. Q0
- b. Q1
- c. Q2
- d. Q3

The correct answer is: Q2

64. Identify the correct definition of lexical analysis.

- a. Breaking a sequence of characters into packets.
- b. Breaking a sequence of characters into groups.
- c. Breaking a sequence of characters into tokens.
- d. Breaking a sequence of characters into lines.

The correct answer is: Breaking a sequence of characters into tokens.

65. Which of the following is used to store user data on the server-side in web applications?

- a. Cookies
- b. Sessions
- c. Local storage
- d. IndexedDB

The correct answer is: Sessions

66. What is the objective of a Serializable schedule?

- a. Find a Non-serial schedule that allows the interleaving of a transaction and produces the same result as a serial schedule.
- b. Find a Serial schedule that allows the interleaving of a transaction and produces the same result as a serial schedule.
- c. Find a Serial schedule that doesn't allow the interleaving of a transaction and produces the same result as a serial schedule.
- d. Find a Non-serial schedule that doesn't allow the interleaving of a transaction and produces the same result as a serial schedule.

The correct answer is: Find a Non-serial schedule that allows the interleaving of a transaction and produces the same result as a serial schedule.

67. Suppose we want to determine the efficiency of an algorithm. How can we measure the space factor?

- a. Count the maximum memory required by the algorithm.
- b. Count the minimum memory required by the algorithm.
- c. Count the average memory required by the algorithm.
- d. Count the maximum disk space needed by the algorithm.

The correct answer is: Count the maximum memory required by the algorithm.

68. All are properties of a substitution cipher except:

- a. The identity of the letters is changed in the encryption process.
- b. The position of the letters is changed in the encryption process.
- c. Uses the same key for the encryption and decryption process.
- d. It has a secure key distribution problem.

The correct answer is: The position of the letters is changed in the encryption process.

69. Which of the following registers is used to hold the address of the instruction?

- a. Accumulator
- b. Instruction register
- c. Address register
- d. Program counter

The correct answer is: Program counter.

70. Which of the following is not true?

- a. Object Identifier is immutable.
- b

. Type constructor is used to create a new object.

- c. Tuple is a collection of records.
- d. The atom constructor is a multi-value constructor.

The correct answer is: The atom constructor is a multi-value constructor.

71. Which of the following logic gates has an output value that is identical to its input?

- a. NAND
- b. NOR
- c. Exclusive-OR
- d. Buffer

The correct answer is: Buffer.

72. For an effective operating system, when should we check for deadlock?

- a. Every time a resource request is made at fixed time intervals.
- b. At fixed time intervals.
- c. Every time a resource request is made.
- d. None of the mentioned.

The correct answer is: Every time a resource request is made at fixed time intervals.

73. In general, which type of environment is the most challenging for agents?

- a. Fully observable, discrete, episodic, static.
- b. Static, episodic, continuous, partially observable.
- c. Partially observable, non-deterministic, non-episodic, dynamic, continuous.
- d. Deterministic, sequential (non-episodic), dynamic, continuous.

The correct answer is: Partially observable, non-deterministic, non-episodic, dynamic, continuous.

74. Finite automaton recognizes:

- a. Any class of language.
- b. Context-free language.
- c. Context-sensitive languages.
- d. Regular languages.

The correct answer is: Regular languages.

75. What does a top-down parser generate?

- a. Rightmost derivation in reverse.
- b. Rightmost derivation.
- c. Leftmost derivation in reverse.

d. Leftmost derivation.

The correct answer is: Leftmost derivation.

76. Consider the following sample code. What would the following function yield when called?

```
```java
public void test(boolean a, boolean b) {
 if (a) {
 System.out.println("A");
 } else if (a && b) {
 System.out.println("A && B");
 } else {
 if (!b) {
 System.out.println("!B");
 } else {
 System.out.println("None");
 }
 }
}
```
```

- a. If `a` and `b` are both true, then the output is "A && B".
- b. If `a` is true and `b` is false, then the output is "!B".
- c. If `a` is false and `b` is true, then the output is "None".
- d. If `a` and `b` are both false, then the output is "None".

The correct answer is: If `a` is false and `b` is true, then the output is "None".

77. Which of the following is a valid variable name?

- a. T-3
- b. GGGGG
- c. float
- d. 4thInput

The correct answer is: GGGGG.

78. Consider the following C program:

```
```c
int min() { /* line 1 */
 int I, N; /* line 2 */
 fro ((I = 0, I; /* line 3 */
}
```
```

While creating the object module, what would be the compiler's response about line number 3?

- a. Only syntax error.
- b. No compilation error.
- c. Only lexical error.
- d. Both lexical and syntax error.

The correct answer is: Only syntax error.

79. What is the maximum number of IP addresses that can be assigned to hosts on a local subnet that uses the 255.255.255.224

subnet mask?

- a. 14
- b. 32
- c. 24
- d. 30

The correct answer is: 30.

80. An array is declared and initialized as follows:

```
```java
int A[4][4] = {{1, 2, 3, 4}, {5, 6, 7, 8}, {9, 8, 7, 6}, {5, 4, 3, 2}};
```
```

What are the values of `A[1][1]`, `A[2][3]`, and `A[3][3]`, respectively?

- a. 7, 6, 3
- b. 6, 6, 2
- c. 6, 2, 6
- d. 2, 6, 6

The correct answer is: 6, 6, 2.

81. Which one of the following is a Nonfunctional requirement?

- a. Admin can upload photos
- b. Admin can search book
- c. Librarian can delete student
- d. The system delivers the search items within 2 seconds

The correct answer is: The system delivers the search items within 2 seconds

82. Which of the following is false about functional requirement?

- a. Requirements should complete
- b. Requirement should be accurate as possible.
- c. Two requirements cannot combine into one.
- d. A requirement can be interpreted in many exclusive ways.

The correct answer is: A requirement can be interpreted in many exclusive ways.

83. Which one of the following methods can be overriding?

- a. Abstract method
- b. Final Method

- c. Private method
- d. Static method

The correct answer is: Abstract method

84. What is the maximum number of children that a node can have in a binary tree?

- a. 3
- b. 1
- c. 4
- d. 2

The correct answer is: 2

85. Assume transaction A holds a shared lock R. If transaction B also requests for a shared lock on R

- a. It will result in a deadlock situation
- b. It will be granted as soon as it is released by A
- c. It will immediately be granted
- d. It will immediately be rejected

The correct answer is: It will immediately be granted

86. The operating system is responsible for?

- a. bad-block recovery
- b. booting from disk
- c. disk initialization
- d. all of the mentioned

The correct answer is: all of the mentioned

87. In the dirty read problem, one transaction reads:

- a. an uncommitted value of another transaction
- b. the committed value for another transaction
- c. another transaction
- d. one transaction commits another transaction

The correct answer is: One transaction reads an uncommitted value of another transaction

88. All are properties of the integrity goal of a computer system, except

- a. Ensures there was an authorized modification
- b. Keep resources from active attacks
- c. Integrity can be compromised by Deception
- d. Integrity can be compromised by Disclosure

The correct answer is: Integrity can be compromised by Disclosure

89. Suppose we want to determine the efficiency of the algorithm, then how can we measure the space factor?

- a. To count the maximum memory required by the algorithm
- b. To count the minimum memory required by the algorithm
- c. To count the average memory required by the algorithm
- d. To count the maximum disk space needed by the algorithm

The correct answer is: To count the maximum memory required by the algorithm

90. When does a page fault occur?

- a. The page is present in memory.
- b. Deadlock occurs.
- c. The page does not present in memory.
- d. Buffering occurs.

The correct answer is: The page does not present in memory.

91. Which one of the following is not LAN technologies?

- a. Token Ring
- b. Ethernet
- c. X.25 or Frame Relay
- d. None

The correct answer is: X.25 or Frame Relay

92. All of the following are the roles of a system admin related to the file system and disk space management EXCEPT?

- a. Assigning user ID and User name for system users
- b. Determining disk usage and quota
- c. Creating a Mounting point

d. Partition disk space

The correct answer is: Assigning user ID and User name for system users

93. Which of the following is not true about requirement elicitation?

- a. Errors introduced during requirements elicitation are expensive to correct.
- b. Define the purpose of the system by collaboration of the client and developers.
- c. Requirements elicitation is less challenging than Analysis.
- d. Requirements specification is the result of requirement elicitation.

The correct answer is: Requirements elicitation is less challenging than Analysis

94. Which one of the following grammars is unambiguous?

A.

$E \rightarrow I$

$E \rightarrow E + E$

$E \rightarrow E * E$

$E \rightarrow (E)$

$I \rightarrow \varepsilon \mid 0 \mid 1 \mid 2 \mid \dots \mid 9$

B.

$S \rightarrow aSb \mid SS$

$S \rightarrow \varepsilon$

C.

$E \rightarrow E + T \mid T$

$T \rightarrow T \times F \mid F$

$F \rightarrow id$

D.
 $S \rightarrow A / B$
 $A \rightarrow aAb / ab$
 $B \rightarrow abB / \epsilon$

The correct answer is:

$E \rightarrow E + T / T$
 $T \rightarrow T \times F / F$
 $F \rightarrow id$

95. Which method is used to prevent threads or processes from accessing a single resource?

- a. PCB
- b. Semaphore
- c. Job Scheduler
- d. Non-Contiguous Memory Allocation

The correct answer is: Semaphore

96. Before a program is written, the programmer must ensure to:

- a. Clear and understand what data are to be used
- b. Clear and understand the desired result
- c. Clear and understand the procedure to be used
- d. All of the above

The correct answer is: All of the above

97. Which internet layer address is responsible for providing diagnostic functions and reporting errors or conditions regarding the delivery of IP packets?

- a. Internet Protocol (IP)
- b. Address Resolution Protocol (ARP)
- c. Internet Control Message Protocol (ICMP)
- d. Internet Group Message Protocol (IGMP)

The correct answer is: Internet Control Message Protocol (ICMP)

98. Hana obtains a network address of 172.16.0.0/19 from an internet service provider (ISP) for a small-sized internet café with a limited number of rooms. How many subnets and hosts does Hana need to configure the given address as subnets and hosts?

- a. 7 subnets and 30 valid hosts
- b. 6 valid subnets and 8190 valid hosts in each subnet
- c. 8 subnets and 2046 valid hosts in each subnet
- d. 7 subnets and 2046 valid hosts in each subnet

The correct answer is: 6 valid subnets and 8190 valid hosts in each subnet

99. All are true about the Database except:

- a. Program data independence.
- b. Self-describing nature
- c. Sharing of data and multi-user transaction processing
- d. Support of single views of the data

The correct answer is: Support of single views of the data

100. Which one of the following is a supervised machine learning task?

- a. Summarization
- b. Time series analysis
- c. Sequence discovery
- d. Clustering

The correct answer is: Time series analysis

101. Which of the architecture is power efficient?

- a. RISC
- b. ISA
- c. IANA
- d. CISC

The correct answer is: RISC

102. Choose

the correct output for the following sequence of operations.

- a. 8 5 2 5 1
- b. 8 5 5 2 1
- c. 8 2 5 5 1
- d. 8 1 2 5 5

The correct answer is: 8 5 2 5 1

103. Which one of the following statements is true?

- a. One-time pad cipher is highly secure than Vigenere cipher
- b. Symmetric key cryptography has a slow encryption process than asymmetric key encryption technique
- c. Hill-cipher can be easily breakable by brute force key search
- d. All of the above

The correct answer is: One-time pad cipher is highly secure than Vigenere cipher